ECG Heartbeat Classification: A 1-D Residual CNN Approach with SMOTE-Based Class Imbalance Correction

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***Abstract*—**Automated classification of electrocardiogram (ECG) heartbeats supports cardiovascular monitoring, yet the extreme class imbalance found in real-world arrhythmia datasets biases most learning algorithms toward the dominant Normal beat category. This report presents the design, implementation, and evaluation of a one-dimensional residual convolutional neural network (1-D ResNet) trained on the PhysioNet MIT-BIH Arrhythmia Database to classify heartbeats into the five AAMI EC57 categories (N, S, V, F, Q). Two controlled experiments were carried out under identical architectural and training conditions: a baseline model trained on the raw, imbalanced training set, and a second model trained on a class-balanced training set produced with the Synthetic Minority Over-sampling Technique (SMOTE). Both models were evaluated on the same held-out, imbalanced test set of 21,892 beats. The baseline achieved an overall test accuracy of 98.70%, while the SMOTE-augmented model achieved 98.69% — a negligible difference. Per-class analysis, however, shows that the SMOTE model meaningfully improved recall on the rare and clinically important Supraventricular (+4.67 points) and Fusion (+4.32 points) beat categories, at a modest cost to precision on those same classes. These findings illustrate that overall accuracy alone is a misleading indicator of model quality for imbalanced medical classification tasks.

***Keywords*—**ECG classification, arrhythmia detection, deep learning, 1-D residual CNN, class imbalance, SMOTE, MIT-BIH database.

I. INTRODUCTION

An electrocardiogram (ECG) records the electrical activity of the heart over time, and manual inspection of continuous ECG traces by cardiologists is time-consuming and susceptible to human error. Because cardiovascular disease remains a leading cause of death worldwide, accurate and low-cost automated detection of irregular heartbeats (arrhythmias) is highly desirable. Convolutional neural networks (CNNs) have emerged as an effective tool for this task because they learn discriminative temporal features directly from raw signal windows, removing the need for hand-crafted statistical or wavelet-based features.

The central obstacle in automated heartbeat classification is class imbalance: in the widely used MIT-BIH Arrhythmia Database, Normal beats account for roughly 83% of all recorded beats, while dangerous arrhythmias such as Supraventricular and Fusion beats make up less than 3% combined. A classifier that is optimized purely for overall accuracy can therefore achieve a deceptively high score while almost entirely failing to detect the rare classes that matter most clinically.

This project reproduces a deep residual 1-D CNN architecture and systematically studies whether Synthetic Minority Over-sampling (SMOTE) can improve the model's sensitivity to minority arrhythmia classes without materially harming overall performance. Two models sharing an identical architecture, optimizer, and training schedule were trained — one on the original imbalanced data and one on a SMOTE-balanced version of the same data — and evaluated on an untouched, imbalanced test set to ensure a fair, controlled comparison. The remainder of this report documents the dataset, methodology, experimental results, and clinical implications of the comparison.

II. RELATED WORK

Kachuee et al. proposed a deep residual 1-D CNN for classifying ECG beats according to the AAMI EC57 standard and reported an average accuracy of 93.4% on the MIT-BIH dataset, later transferring the learned representation to a myocardial-infarction classification task on the PTB Diagnostic dataset [1]. Earlier approaches to this problem relied on hand-crafted, statistical or wavelet-domain features paired with conventional classifiers such as support vector machines, decision trees, or random forests, which are generally less scalable and less accurate than end-to-end deep representations.

Residual connections, introduced by He et al. for image recognition, mitigate the vanishing-gradient problem and allow substantially deeper networks to be trained effectively [6]; this project adopts the same principle in one dimension. To address class imbalance, Chawla et al. introduced the Synthetic Minority Over-sampling Technique (SMOTE), which generates new minority-class instances by interpolating between existing neighbours in feature space rather than duplicating them outright, reducing the tendency of a classifier to overfit on repeated samples [5]. This project combines the Kachuee-style 1-D ResNet architecture with SMOTE and quantifies the resulting trade-off between overall accuracy and minority-class sensitivity.

III. DATASET

The PhysioNet MIT-BIH Arrhythmia Database is the standard benchmark for this task. It contains ECG recordings from 47 subjects, each beat annotated by at least two cardiologists. In this project, the pre-processed and beat-segmented release distributed on Kaggle by Fazeli is used: each beat is represented as 187 samples (approximately 1.5 seconds at 125 Hz), min–max normalized to the range [0, 1] and zero-padded to a fixed length. Every beat is labeled with one of five classes defined by the AAMI EC57 standard, summarized in Table I.

TABLE I
CLASS DISTRIBUTION — AAMI EC57 CATEGORIES
(TRAINING SET)

Class	Description	Count	%
N	Normal / bundle-branch block / escape	72,471	82.7%
S	Supraventricular premature beat	2,223	2.5%
V	Premature ventricular contraction	5,788	6.6%
F	Fusion of ventricular and normal	641	0.7%
Q	Paced / unclassifiable beat	6,431	7.3%

The training split contains 87,554 beats and the test split contains 21,892 beats, both drawn from the same five classes. Class N alone accounts for over four-fifths of the training data, while Class F contributes less than one percent — an imbalance ratio of roughly 113:1 between the largest and smallest class, as visualized on a logarithmic scale in Fig. 1.

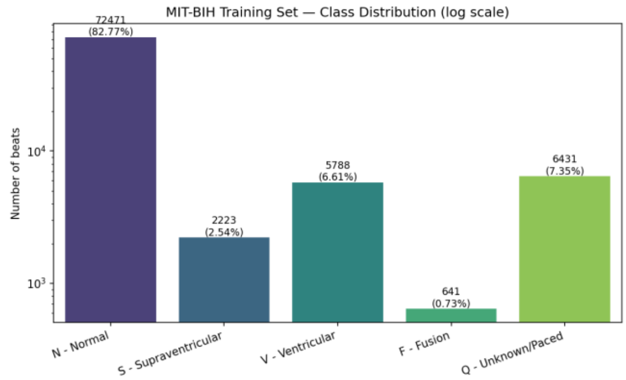

Fig. 2. MIT-BIH training-set class distribution shown on a logarithmic scale, highlighting the dominance of Class N.

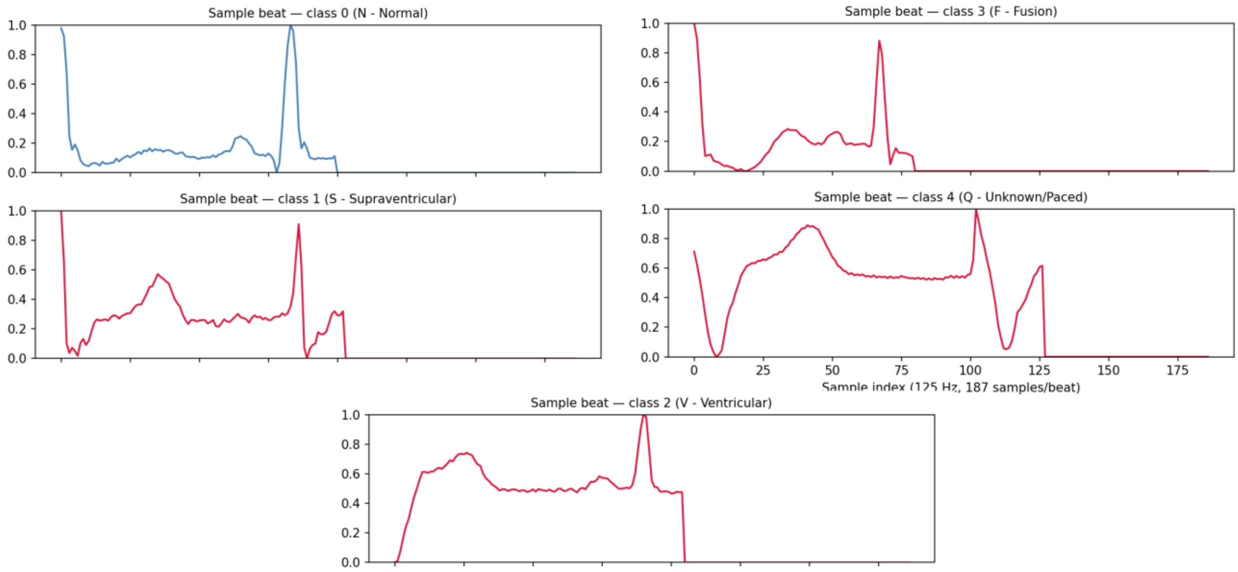

Fig. 1. One representative beat waveform from each of the five AAMI classes. Morphological differences such as the widened, notched complex of Class F and the irregular pre-R-peak segment of Class S distinguish arrhythmic beats from the clean Class N morphology.

IV. METHODOLOGY

The overall experimental workflow, illustrated by the source files data.py and model.py, consists of four stages: data loading and reshaping, network construction, (optionally) SMOTE balancing, and supervised training with early stopping.

A. Data Preprocessing

Each CSV file is loaded with pandas; columns 0–186 form the signal and column 187 the integer class label. The 2-D feature matrix is reshaped to a 3-D tensor of shape (samples, 187, 1) so that the trailing channel dimension is compatible with Keras Conv1D layers. Training samples are shuffled with a fixed random seed (42) prior to fitting so that the validation split drawn by Keras contains a representative mix of classes.

B. Network Architecture

The network follows the residual 1-D CNN design of Kachuee et al. [1]. An initial Conv1D layer (32 filters, kernel size 5) is followed by five identical residual blocks. Each block applies two Conv1D(32, 5) layers separated by a ReLU activation, adds the block input back to the output through a skip connection, applies a further ReLU, and down-samples with a MaxPool1D layer (pool size 5, stride 2). The resulting feature map is flattened and passed through two fully-connected layers of 32 ReLU units before a final 5-way softmax layer. In total the network comprises 13 weight layers — 11 convolutional and 2 dense hidden layers plus the output layer.

C. Handling Class Imbalance with SMOTE

SMOTE cannot operate directly on the 3-D tensors used by the CNN, so the training tensor is temporarily flattened to two dimensions before oversampling and reshaped back to (samples, 187, 1) afterward. Critically, SMOTE (random_state = 42) is applied exclusively to the training set; the test set is left completely untouched so that evaluation always reflects real, unmodified clinical data and no synthetic sample can leak into the reported metrics. After balancing, all five classes contain an equal number of training beats, removing the numerical dominance of Class N seen in Fig. 1.

D. Training Configuration

To isolate the balancing technique as the sole independent variable, the baseline and SMOTE models share an identical architecture and training configuration, summarized in Table II. Both models were compiled with the Adam optimizer and sparse categorical cross-entropy loss, and trained for up to 30 epochs with EarlyStopping (patience 5, restoring the best weights) and ReduceLROnPlateau (patience 3, factor 0.5) callbacks monitoring the validation loss.

TABLE II
TRAINING HYPERPARAMETERS

Hyperparameter	Value
Optimizer	Adam
Learning rate	1e-3
Loss function	Sparse categorical cross-entropy
Batch size	256
Max epochs	30
Validation split	10%
EarlyStopping patience	5
ReduceLROnPlateau	patience 3, factor 0.5
Input / output shape	(187, 1) → 5 classes

V. EXPERIMENTAL RESULTS

Both trained models were evaluated on the identical, untouched 21,892-beat test set using scikit-learn's classification_report and confusion_matrix utilities, so that any performance difference can be attributed solely to the training-data balancing strategy.

A. Baseline Model Performance

The baseline model reached an overall test accuracy of 98.70% (macro-F1 92.78%, weighted-F1 98.67%). As shown in Fig. 3, the confusion matrix is dominated by the diagonal for Class N, whose recall and precision both exceed 98.9%. However, recall on the minority classes is markedly weaker: only 79.32% of true Class S beats and 78.40% of true Class F beats are correctly identified, with most of the remaining beats misclassified as Class N — precisely the failure mode that overall accuracy conceals.

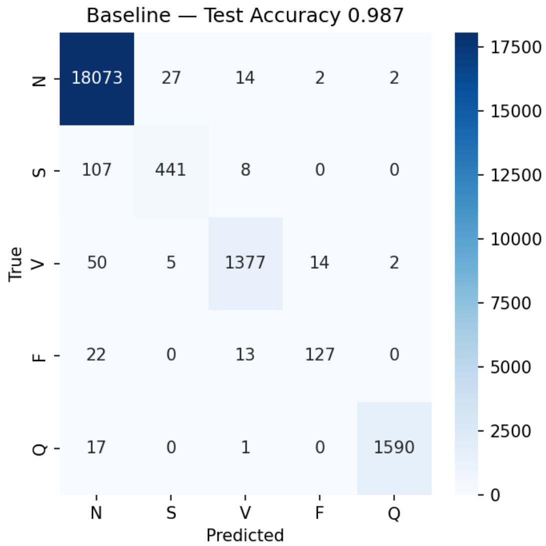

Fig. 3. Baseline model confusion matrix on the 21,892-beat test set (test accuracy 0.987).

B. SMOTE Model Performance

The SMOTE-trained model reached an almost identical overall accuracy of 98.69% (macro-F1 92.82%, weighted-F1 98.68%), as shown in Fig. 4. Despite the near-unchanged aggregate accuracy, recall on Class S rose to 83.99% (+4.67 points) and recall on Class F rose to 82.72% (+4.32 points). This improvement came at the cost of lower precision for the same two classes — Class S precision fell from 93.23% to 87.78%, and Class F precision fell from 88.81% to 83.23% — because the SMOTE-trained model is more willing to flag borderline beats as arrhythmic.

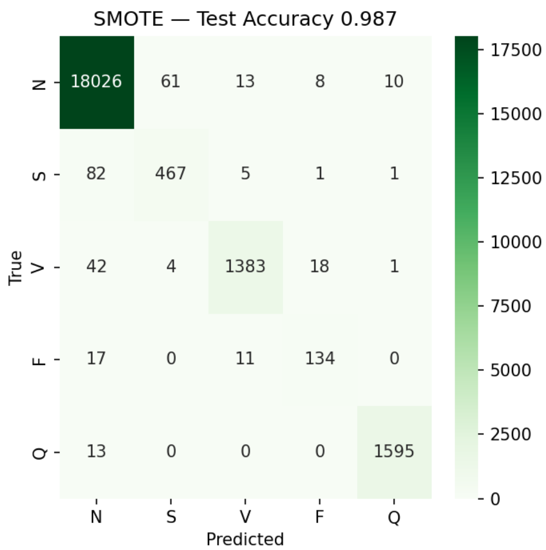

Fig. 4. SMOTE-trained model confusion matrix on the same test set (test accuracy 0.987).

C. Comparative Analysis

Table III summarizes precision, recall, and F1-score for both models across all five classes on the shared test set (support = 21,892). The two confusion matrices are compared directly in Fig. 5. Classes V and Q, which already had moderate-to-large support and reasonably good baseline recall, show only marginal changes after SMOTE (V recall +0.41 points, Q recall +0.31 points), while the smallest classes, S and F, show the largest recall gains — consistent with SMOTE's purpose of strengthening decision boundaries for severely under-represented categories.

TABLE III
PER-CLASS PRECISION / RECALL / F1 (%) — BASELINE VS. SMOTE

Class	Prec. (Base)	Prec. (SMOTE)	Rec. (Base)	Rec. (SMOTE)	F1 (Base)	F1 (SMOTE)	Support
N	98.93	99.15	99.75	99.49	99.34	99.32	18,118
S	93.23	87.78	79.32	83.99	85.71	85.85	556
V	97.45	97.95	95.10	95.51	96.26	96.71	1,448
F	88.81	83.23	78.40	82.72	83.28	82.97	162
Q	99.75	99.25	98.88	99.19	99.31	99.22	1,608

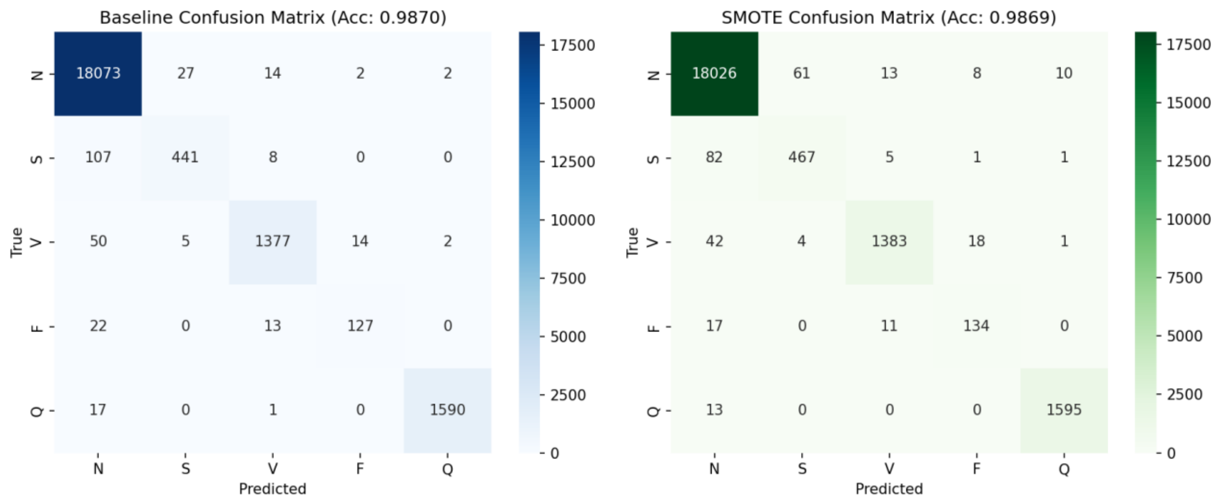

Fig. 5. Side-by-side confusion matrices for the baseline (left) and SMOTE-trained (right) models on the same 21,892-beat test set. The SMOTE matrix shows visibly fewer S and F beats collapsing into the Class N column.

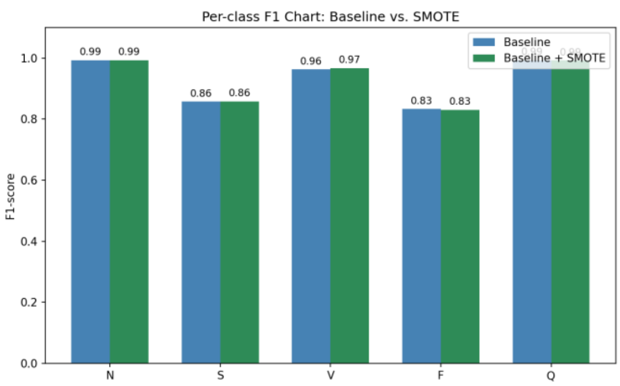

Fig. 6. Per-class F1-score for the baseline versus SMOTE-trained model. The largest gains occur for Class V, while Classes N and Q are essentially unchanged.

VI. DISCUSSION

The results illustrate a classic precision–recall trade-off. By exposing the network to a balanced training distribution, SMOTE shifts the decision boundary to be more sensitive toward arrhythmic beats. In a clinical workflow this trade is usually desirable: a false positive (flagging a normal beat as suspicious) merely prompts a clinician to double-check the trace, whereas a false negative (missing a true arrhythmia) could delay treatment of a dangerous condition. On this basis, the SMOTE model can be considered clinically preferable even though its raw accuracy is fractionally lower.

Overall accuracy remained essentially flat (98.70% → 98.69%) purely because Class N constitutes 82.7% of the test set: a tiny fractional change in Class N performance is enough to mask substantial four-to-five-point recall improvements on the rare classes. This confirms that accuracy alone is a poor metric for imbalanced medical classification, and that macro-averaged or per-class recall figures should be reported alongside it.

VII. LIMITATIONS

Several limitations should be considered when interpreting these results:

- Single-lead input — the model uses only ECG Lead II; 12-lead ECGs used in clinical practice provide additional spatial information not captured here.
- No patient-wise split verification — the standard train/test partition does not explicitly guarantee that beats from the same patient are isolated to one split, which could inflate reported accuracy.

- Physiological plausibility of SMOTE — SMOTE performs linear interpolation in a high-dimensional signal space and may generate synthetic waveforms that do not correspond to biologically realistic morphologies.
- No uncertainty estimation — the softmax output provides a point probability with no principled confidence or calibration measure, which is important for clinical decision support.
- Single architecture — only the Kachuee-style 1-D ResNet was evaluated; comparisons against transformers, LSTMs, or classical baselines were out of scope.
- No explainability analysis — post-hoc interpretability methods (e.g., Grad-CAM, SHAP) were not applied to identify which waveform segments drive each prediction.

VIII. CONCLUSION AND FUTURE WORK

This project successfully reproduced a deep residual 1-D CNN for ECG heartbeat classification and used it as a controlled testbed for studying SMOTE-based class-imbalance correction. The baseline model achieved 98.70% test accuracy, exceeding the 93.4% average accuracy reported by Kachuee et al. [1] — a difference plausibly attributable to differences in hyperparameters, early-stopping configuration, or evaluation protocol rather than to the underlying architecture itself. Applying SMOTE to the training data alone left overall accuracy virtually unchanged (98.69%) while improving recall on the two rarest and most clinically significant arrhythmia classes, Supraventricular (+4.67 points) and Fusion (+4.32 points) beats.

Future work could replace SMOTE with class-weighted loss or focal loss to address imbalance without synthesizing data, explore GAN- or VAE-based augmentation for more physiologically realistic minority samples, evaluate attention-based or transformer architectures for long-range temporal dependencies, enforce patient-wise cross-validation to rule out data leakage, and apply post-hoc explainability methods to verify that the network attends to clinically meaningful segments such as the QRS complex.

ACKNOWLEDGMENT

The authors would like to thank Md Mayenul Islam and the Department of Electrical and Electronic Engineering, [Name of Institution], for their guidance and support throughout this project.

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